

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5112 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 5

TOTAL MARKS: 25

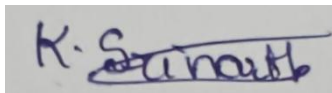
Annexure A CONSTRAINT-BASED PROBLEM SOLVING

Code of Conduct:

I ____K.Srinath(192524220)_____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'K. Srinath' written in a cursive style.

1. 1. INTRUSION DETECTION DEPLOYMENT UNDER REAL-TIME CONSTRAINTS

A financial organization plans to deploy an Intrusion Detection System (IDS) to analyze high-volume traffic in real time while using limited shared CPU resources.

Given Constraints

Latency ≤ 5 m per packet

Detection accuracy $\geq 96\%$

False positive rate $\leq 1.5\%$

Traffic load = 150,000 packets/second

CPU resources are limited and shared

PROPOSED SOLUTION

Deploy a hybrid IDS. Use a fast packet-filtering stage followed by optimized signature-based detection for known attacks. Only suspicious or unmatched traffic is sent to a lightweight anomaly detector. This reduces the amount of expensive analysis while retaining the ability to detect unusual and previously unseen behavior.

CONSTRAINT ANALYSIS

Latency

Fast filtering, efficient signature matching, and selective anomaly analysis keep the real-time path lightweight.

Accuracy

Combining signatures for known attacks with anomaly detection for suspicious behavior provides broader detection coverage.

False positives

Use high-confidence signatures, trusted traffic baselines, and tuned anomaly thresholds.

Traffic load

Process all packets through lightweight filtering, but reserve deep inspection for suspicious flows.

CPU

Remove obsolete rules and avoid expensive analysis of every packet.

Trade-off Analysis

Approach	Benefit	Trade-off
Signature-based	Fast and accurate for known attacks	Cannot reliably identify new attacks
Anomaly-based	Can detect unknown behavior	Higher CPU use and possible false positives
Hybrid	Balances coverage and performance	Requires careful tuning

Constraint Satisfaction

Requirement	Proposed Approach	How It Is Addressed
Latency ≤ 5 ms per packet	Fast filtering + selective analysis	Keeps the real-time path lightweight.
Detection accuracy $\geq 96\%$	Hybrid/curated high-confidence detection	Provides broad detection coverage.
False positive rate $\leq 1.5\%$	Threshold tuning + high-confidence rules	Reduces unnecessary alerts.
Traffic load = 150,000 packets/second	Lightweight first-stage processing	Handles high traffic efficiently.
CPU resources are limited and shared	Selective monitoring/analysis	Limits resource consumption.

Final Justification

The hybrid IDS is the most suitable solution because it balances real-time performance and detection capability. Signature detection handles known attacks quickly, while anomaly analysis adds protection against suspicious and unknown behavior. With filtering, rule optimization, threshold tuning, and selective deep inspection, the design is intended to satisfy the 5 ms latency, 96% accuracy, 1.5% false-positive, 150,000 packets/second, and limited-CPU constraints.

2. WORM CONTAINMENT STRATEGY UNDER NETWORK AVAILABILITY CONSTRAINTS

An enterprise network faces rapid worm propagation. The solution must contain the worm quickly while keeping network downtime very low across multiple locations with limited centralized control.

Given Constraints

Network downtime $\leq 1\%$

Detection and containment within 2 minutes

Systems distributed across multiple locations

Limited centralized control

Proposed Solution

Use a distributed containment architecture with IDS/IPS, firewalls, and network segmentation. Local IDS sensors detect worm-like scanning and abnormal traffic. IPS blocks high-confidence malicious traffic, firewalls install temporary blocking rules, and segmentation isolates affected hosts or VLANs instead of shutting down entire sites.

Constraint Analysis

Detection and containment

Local IDS/IPS sensors and automated firewall responses reduce delay.

Availability

Isolate affected hosts or segments rather than taking down complete networks.

Multiple locations

Each site has local detection and containment capability.

Limited central control

Predefined local policies allow automated response without waiting for a central administrator.

Trade-off Analysis

Approach	Benefit	Trade-off
Automated isolation	Stops spread rapidly	Overly aggressive blocking can affect legitimate traffic
Segmentation	Limits lateral movement	Requires properly designed network zones
Distributed response	Works despite limited central control	Policies must remain consistent

Constraint Satisfaction

Requirement	Proposed Approach	How It Is Addressed
Network downtime $\leq 1\%$	Selective host/VLAN isolation	Protects unaffected services.
Detection and containment within 2 minutes	Local IDS/IPS + automated response	Reduces response delay.
Systems distributed across multiple locations	Distributed controls	Enables local containment.
Limited centralized control	Preconfigured local policies	Reduces central dependency.

Final Justification

A distributed strategy minimizes worm propagation while preserving business continuity. High-confidence malicious traffic can be blocked automatically, while only affected hosts or segments are isolated. This minimizes downtime and supports the two-minute containment requirement.

3. FIREWALL RULE OPTIMIZATION UNDER PERFORMANCE CONSTRAINTS

A company firewall contains more than 10,000 rules and suffers performance degradation. The rule base must be optimized without adding hardware or weakening strict access-control policies.

Given Constraints

Throughput $\geq$ 800 Mbps

Rule processing time must be minimized

Strict access control must be maintained

No additional hardware

Proposed Solution

Optimize the rule set using rule aggregation, prioritization, redundancy elimination, and careful ordering. Combine compatible rules, remove duplicate or shadowed rules, eliminate obsolete entries, and place frequently matched rules efficiently while preserving explicit security restrictions.

Constraint Analysis

Rule aggregation

Combine compatible rules to reduce the number of evaluations.

Prioritization

Place commonly matched and security-critical rules efficiently.

Redundancy elimination

Remove duplicate, shadowed, and obsolete rules.

Security preservation

Validate every change against the original access-control policy.

Trade-off Analysis

Technique	Benefit	Trade-off
Aggregation	Fewer rules and comparisons	Incorrect merging can broaden access
Prioritization	Lower average processing time	Needs traffic-pattern analysis
Redundancy removal	Simpler rule base	Requires careful dependency checking

Constraint Satisfaction

Requirement	Proposed Approach	How It Is Addressed
Throughput $\geq$ 800 Mbps	Rule optimization	Reduces rule-processing overhead.
Rule processing time must be minimized	Ordering + redundancy removal	Reduces comparisons.
Strict access control must be maintained	Policy-preserving validation	Maintains strict control.
No additional hardware	Software/rule optimization	Requires no additional hardware.

Final Justification

Rule optimization improves firewall performance without additional hardware. Aggregation, efficient ordering, and removal of redundant rules reduce processing overhead. Careful validation preserves the original security policy, making the optimized rule set suitable for achieving the required throughput.

4. IPS DEPLOYMENT UNDER RESOURCE AND ACCURACY CONSTRAINTS

An organization needs an IPS that blocks most known attacks while operating within a 1 GB signature-memory limit and supporting a 500 Mbps network.

Given Constraints

Block $\geq 90\%$ of known attacks

False positive rate $\leq 3\%$

Signature memory ≤ 1 GB

Network speed = 500 Mbps

Proposed Solution

Use a curated, high-confidence signature set containing the most relevant and high-impact attack signatures. Remove obsolete and duplicate signatures. Apply efficient traffic filtering and selective deep inspection. Use adaptive updates to replace outdated signatures and continuously tune noisy rules.

Constraint Analysis

Attack blocking

Prioritize high-coverage signatures for common and severe attacks.

False positives

Tune thresholds and disable noisy low-confidence signatures.

Memory

Keep only relevant signatures and remove duplicates/obsolete entries.

500 Mbps

Use efficient filtering and avoid unnecessary deep inspection.

Trade-off Analysis

Factor	Benefit	Trade-off
Curated signatures	High security value within memory	Rare attacks may receive less coverage
Adaptive updates	Keeps protection current	Requires monitoring and testing
Selective inspection	Reduces processing overhead	Needs good traffic classification

Constraint Satisfaction

Requirement	Proposed Approach	How It Is Addressed
Block $\geq 90\%$ of known attacks	High-coverage curated signatures	Prioritizes known attacks.
False positive rate $\leq 3\%$	Threshold tuning + high-confidence rules	Reduces unnecessary alerts.
Signature memory ≤ 1 GB	Remove obsolete/duplicate signatures	Controls memory usage.
Network speed = 500 Mbps	Efficient filtering	Supports network performance.

Final Justification

The resource-aware IPS focuses limited memory and processing on high-value signatures. Efficient filtering supports the 500 Mbps requirement, while signature tuning keeps false positives within the limit. Adaptive updates maintain protection without uncontrolled signature growth. The strategy therefore balances attack coverage, accuracy, memory, and performance.

5. MALWARE DETECTION USING BEHAVIORAL ANALYSIS UNDER SYSTEM CONSTRAINTS

A company wants lightweight behavioral malware detection that identifies unknown threats quickly while keeping endpoint CPU usage and storage consumption low.

Given Constraints

Endpoint CPU usage $\leq 15\%$

Detection within 10 seconds of execution

Must detect unknown/zero-day threats

Storage for logs and analysis is limited

Proposed Solution

Monitor a small set of high-value behavioral indicators such as process creation, suspicious child processes, persistence attempts, unusual file modifications, abnormal network connections, and privilege-related behavior. Score events locally in real time. Store compact summaries for normal activity and detailed records only for suspicious sequences.

Constraint Analysis

CPU

Selective event monitoring and lightweight scoring reduce endpoint overhead.

Detection speed

Real-time collection and immediate behavioral scoring support detection within 10 seconds.

Unknown threats

Behavioral indicators do not depend only on known malware signatures.

Storage

Compact summaries and detailed logging only for suspicious events conserve storage.

Trade-off Analysis

Factor	Benefit	Trade-off
Real-time monitoring	Fast detection	Continuous monitoring consumes some CPU
Behavioral correlation	Can identify unknown threats	Thresholds require tuning
Selective logging	Reduces storage	Less history for benign activity

Constraint Satisfaction

Requirement	Proposed Approach	How It Is Addressed
Endpoint CPU usage $\leq 15\%$	Selective monitoring/analysis	Limits resource consumption.
Detection within 10 seconds of execution	Real-time scoring	Supports rapid detection.
Must detect unknown/zero-day threats	Behavior-based detection	Does not depend only on known signatures.
Storage for logs and analysis is limited	Compact and selective logs	Controls storage use.

Final Justification

A lightweight behavioral system provides the best balance between speed, accuracy, CPU use, and storage. It can detect suspicious behavior even when no known signature exists. Selective monitoring and logging keep resource usage controlled while real-time scoring supports the 10-second detection requirement.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyses constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided